



JURONG JUNIOR COLLEGE

JC2 Preliminary Examination 2018

CANDIDATE
NAME

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CLASS

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INDEX
NUMBER

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BIOLOGY

9744/04

Paper 4 Practical

14 August 2018

Candidates answer on the Question Paper

2 hours 30 minutes

Additional Materials: As listed in the Confidential Instruction

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
Total	

This document consists of **20** printed pages.

Answer **all** questions.

- 1** Protease enzymes catalyse the hydrolysis of proteins into smaller, more soluble molecules.

In this investigation, the substrate for protease is the protein casein found in milk.

When the enzyme and milk are mixed, the casein is hydrolysed and the mixture will change from cloudy to clear.

You will record this change using the ten-point scale shown.

10	9	8	7	6	5	4	3	2	1	0
cloudy  clear										

You are required to investigate the effect of the enzyme concentration on the hydrolysis of casein.

You are provided with the following:

- 60 cm³ of milk containing casein, labelled **M**,
- 15 cm³ of protease solution, labelled **E1**,
- 15 cm³ of protease solution, labelled **E2**,
- 15 cm³ of protease solution, labelled **E3**,
- 30 cm³ of distilled water, labelled **W**.

Solutions **E1**, **E2** and **E3** are different concentrations of protease.

Protease is harmful. You are advised to wear the eye protection and gloves provided, and wash off any spillages.

Proceed as follows.

Stage 1

Use the beaker or container provided to make a water-bath with warm water, at a temperature between 35 °C and 40 °C.

Maintain the temperature of the water-bath throughout the procedure.

Read through **stages 2–4** and prepare a table to record your results in **stage 5** before proceeding further.

Stage 2

Label five boiling tubes, **B1**, **B2**, **B3**, **B4** and **B5**.

Put 5.0 cm³ of **M** into each of the five boiling tubes.

- (a) Using the ten-point scale, record the initial appearance of the mixtures in the boiling tubes in the table you have prepared in **stage 5**. [1]

Stage 3

This stage must be completed quickly since the reaction will start as soon as protease is added.

Put 1.0 cm³ of the protease solutions, **E1**, **E2** and **E3** or distilled water, **W**, into each boiling tube, as shown in Table 1.1.

Table 1.1

B1	B2	B3	B4	B5
1.0cm ³ of E1	1.0cm ³ of E2	1.0cm ³ of E3	1.0cm ³ of W	1.0cm ³ of E3

Gently mix the solution in each boiling tube, taking care to avoid splashes. Put the boiling tubes into the water-bath and immediately start a stopwatch.

Stage 4

Every 2 minutes, use the ten-point scale to record the appearance of the mixtures in boiling tubes **B1**, **B2**, **B3**, **B4** and **B5** in the table you have prepared in **stage 5**. After 10 minutes, stop recording.

Stage 5

- (b) Use this space to draw your table of results. [5]

(c) Explain the purpose of including boiling tubes **B4** and **B5** in the investigation. [2]

(d) Identify **one** significant source of error in using the ten-point scale to record the appearance of the mixtures. [1]

(e) Describe **three** modifications to your investigation which would improve confidence in your results. [3]

Turn over for the remainder of Question 1

Marshmallows are made using the protein gelatin, which is hydrolysed by a protease enzyme.

Students investigated the activity of protease at different pH values, starting with pH 1.0.

10 marshmallows were arranged in a line and their combined length was measured as shown in Fig. 1.1.

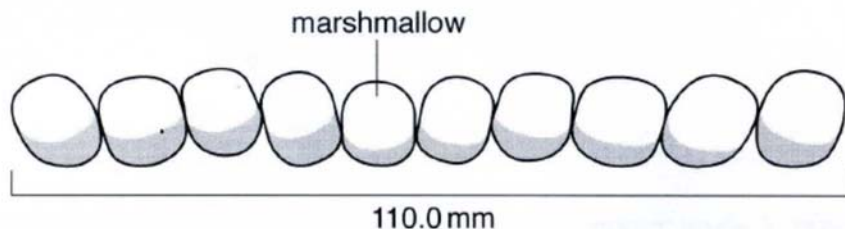


Fig. 1.1

The marshmallows were placed in a Petri dish and covered with a solution of protease maintained at pH 1.0.

After 60 minutes the marshmallows were measured again, as shown in Fig. 1.2, and the decrease in their combined length calculated.

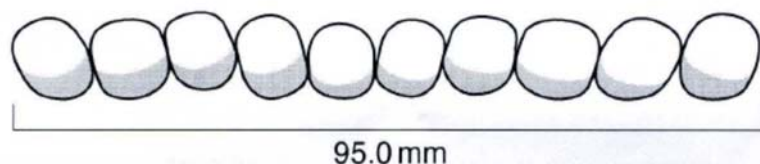


Fig. 1.2

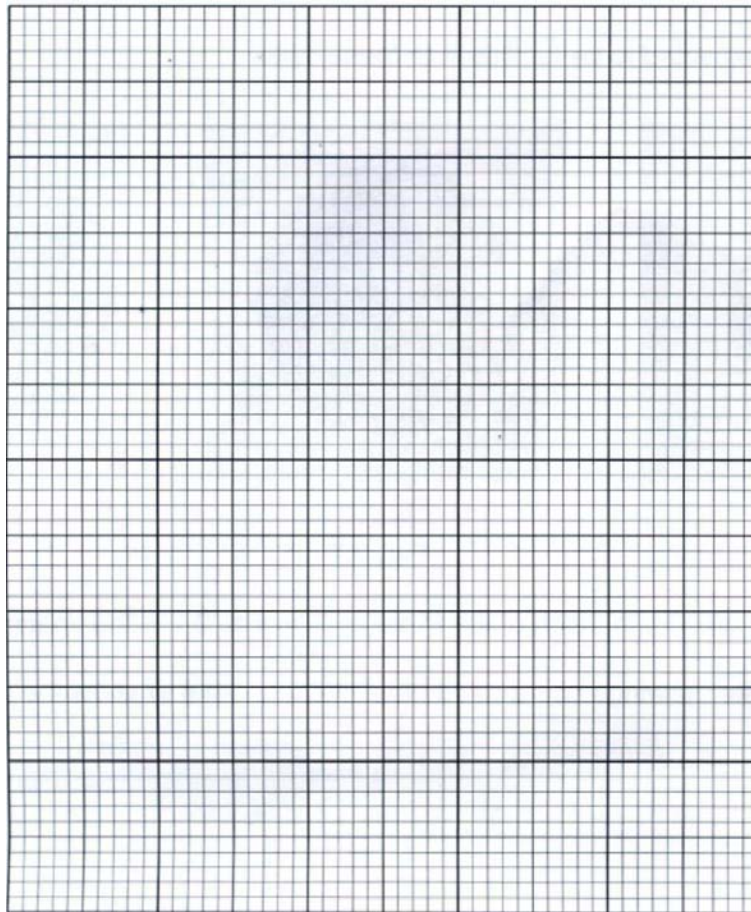
The procedure was repeated with four more sets of marshmallows and solutions of protease maintained at pH 1.5, 3.0, 4.0 and 5.0.

The results of the students' investigation are shown in Table 1.2.

Table 1.2

pH	decrease in combined length of marshmallows /mm
1.0	15.0
1.5	12.0
3.0	8.5
4.0	7.0
5.0	6.0

- (f) Plot a graph of the data shown in Table 1.2. [4]



- (g) (i) Use your graph to estimate the decrease in combined length of marshmallows at pH 3.5. [2]

Show on your graph how you estimated the decrease in length.

decrease in length = [2]

- (ii) Explain the reasons for the differences in the decreases in length at low pH (between 1 and 2) and at higher pH (between 3 and 5). [3]

[Total: 21]

- 2** During this question you will require access to a microscope and slide **B1**.

Slide **B1** shows white blood cells and red blood cells. You are not expected to be familiar with these cells.

- (a)** Examine the slide under the high-power objective lens of your microscope and locate the white blood cells.

Make a detailed drawing of **two** white blood cells in the spaces provided.

The drawings should show differences in appearances between each type of white blood cell.

On one of your drawings, use a ruled label line and label to show the position of the nuclear membrane. [5]

- (b) (i) Using the objective lens selected in (a), complete Table 2.1 so that in (b)(ii) the **mean** number of red blood cells in the field of view can be calculated.

Count every red blood cell for which the **entire cell** is visible in the field of view. [1]

Table 2.1

number of red blood cells in a field of view

- (ii) Calculate the mean number of red blood cells per field of view, using your data from Table 2.1.

Show your working.

mean number of red blood cells per field of view = [2]

(c) Focus the objective lens that you used for counting red blood cells on the stage micrometer provided.

(i) Measure the radius of the field of view and record this in mm.

radius, r , of field of view = mm [1]

(ii) Calculate the area of the field of view and record this area in mm^2 .

Area of circle = πr^2 , given $\pi = 3.142$

Area of field of view = mm^2 [1]

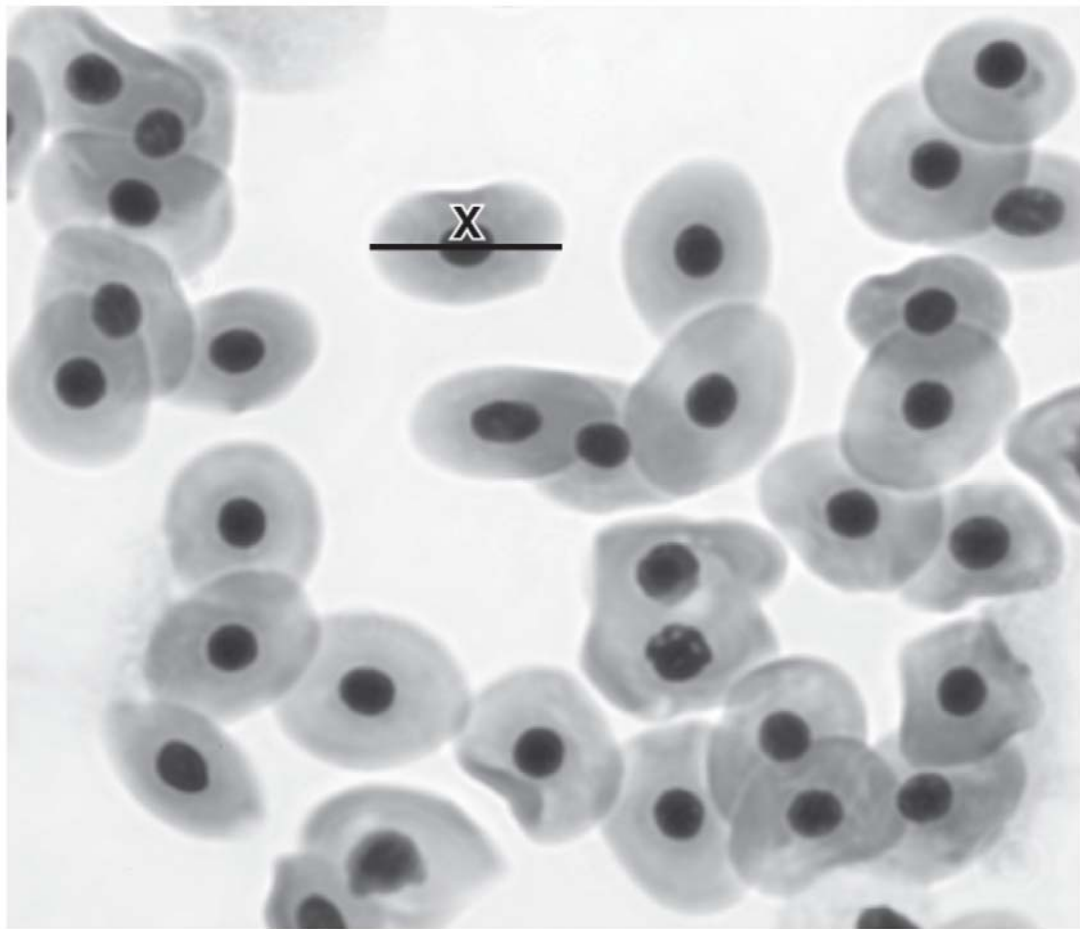
(d) Using your results from (b)(ii) and (c)(ii), calculate the mean density of red blood cells per mm^2 .

mean density of red blood cells = mm^{-2} [1]

Turn over for the remainder of Question 2

Fig. 2.2 is a photomicrograph of blood taken from another type of organism.

You are not expected to be familiar with these cells.



magnification $\times 700$

Fig. 2.2

- (e) Prepare the space below so that it is suitable for you to compare and contrast the blood cells in Fig. 2.2 and the blood cells drawn in (a). [5]

- (f) Calculate the actual diameter of the cell shown by the line **X** in Fig. 2.2.

Show all the steps in your calculation. [2]

..... μm

[Total: 18]

- 3 The information on the label of a bottle of one brand of cleaning solution for contact lenses states that it contains enzymes which remove protein from contact lenses.

The contents of the solution are:

1. Subtilisin A, a protease with an optimum temperature of 60 °C and an optimum pH of 7.0–7.5.
2. Buffered saline, a solution with the same 'balance of ions' as body fluids.
3. Disodium EDTA, a preservative.

A student found a web site that gave the concentration of subtilisin A in different brands of contact lens cleaning solutions. These range between 20 μgcm^{-3} and 100 μgcm^{-3} .

The student simulated a dirty contact lens using the protein gelatin and a thin transparent plastic sheet. One side of the plastic sheet was dipped into melted gelatin containing a dye. The gelatin was then allowed to set.

Fig. 3.1 shows a simulated contact lens.

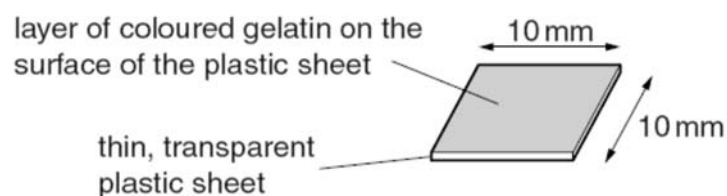


Fig. 3.1

The student tested the activity of subtilisin A by recording the time taken for the coloured gelatin to be removed, leaving the transparent plastic sheet.

Fig. 3.2 shows the apparatus the student used.

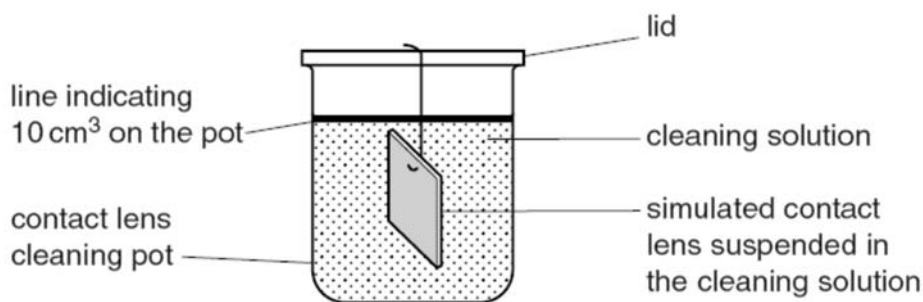


Fig. 3.2

- (a) (i) On Fig. 3.3, add labels for the axes, including units, and show the expected shape of the curve for the factors investigated. [2]

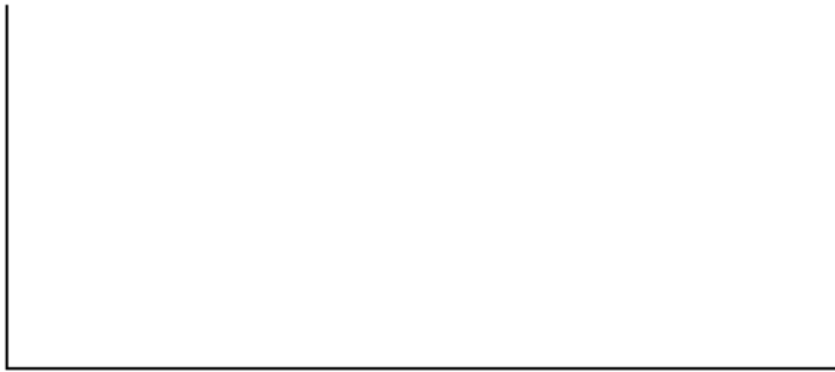


Fig. 3.3

- (ii) Describe how you would use the graph in (i) to find the actual concentration of subtilisin A in the contact lens cleaning solution. [1]

(b) In addition to your answer in (a), design an experiment to find the actual concentration of the protease present in the contact lens cleaning solution.

In your plan, you must use:

- 100 μgcm^{-3} subtilisin A solution
- buffered saline solution
- coloured gelatin on 10 mm x 10 mm plastic sheet shown in Fig. 3.1
- stopwatch
- water bath
- cleaning pot shown in Fig. 3.2

You may select from the following apparatus and plan to use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.
- syringes
- filter funnels

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible
- include layout of results tables with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[13]

[Total: 16]